MLSS Application Statement: Technical Journey in AI/ML

My technical journey in AI/ML started with curiosity and trial and error. I began by learning Python and basic machine learning, then moved toward NLP because I was more interested in systems that can interact with people than in benchmark-only models. Over time, I shifted from running notebooks to building complete, reusable pipelines.

The strongest project I have built so far is AmazonQA-RAG. I designed it as an end-to-end system instead of a single model experiment. In Phase 1, I fine-tuned an instruction model on Amazon-style question-answer data using QLoRA so training could run on limited hardware. In Phase 2, I built a retrieval pipeline with sentence-transformer embeddings and FAISS so the model could answer with evidence rather than guess. In Phase 3, I added RAGAS-based evaluation across three variants (base, RAG plus base, and RAG plus fine-tuned) to measure faithfulness, answer relevancy, context precision, and context recall. In Phase 4, I packaged everything into a Streamlit app with model switching, context inspection, and optional per-query diagnostics.

Before this project, I also explored classic ML workflows such as tabular classification, feature engineering, and model validation, but I kept returning to language systems because they have a direct user-facing impact. I experimented with prompt design, embedding-based search, and few-shot setups to understand where simple methods fail and where architecture choices matter. That gradual progression helped me move from â getting a model to runâ to thinking about system behavior, error cases, and user trust.

This journey taught me practical lessons: good prompts are not enough, retrieval quality can dominate generation quality, and evaluation must be built into the workflow from day one. I also learned to care about reproducibility, configuration design, and failure handling, because real systems break in unexpected ways.

At the same time, I am very aware of my gaps. I want deeper understanding of optimization and generalization, stronger intuition for representation learning, and more systematic knowledge of advanced topics such as multimodal learning, efficient fine-tuning at scale, and principled error analysis. I also want to improve how I read research papers and convert them into clean engineering decisions rather than copying recipes.

MLSS is exactly the environment I need for this next step. I am looking for structured depth, strong mentorship, and peers who challenge each other to think rigorously. I can contribute practical experience in shipping applied NLP systems under constraints, and I will benefit from the program’s emphasis on fundamentals, experiments, and research mindset.

What makes me a strong fit is the combination of curiosity and execution. I do not stop at â the model works.â I keep asking why it works, where it fails, and how to make it more trustworthy. I enjoy the full cycle: building, testing, debugging, and explaining decisions clearly.

In short, my goal is to become an engineer who is both implementation-strong and research-

aware. I want to learn from MLSS, contribute actively to the cohort, and carry that learning into real systems that are useful, reliable, and grounded in evidence.